Ericsson PE Node Troubleshooting SOP: QoS Congestion Event (WFD-PE-02)

Node: WFD-PE-02 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_9262

Description

This Standard Operating Procedure outlines the steps to resolve major QoS congestion events on the Ericsson PE node WFD-PE-02 at the Whitefield POP, caused by microbursts, interface tail drops, and QoS queue starvation. The Ericsson PE node, as an essential component of the provider edge (PE) role, is responsible for directing data traffic between the provider edge and the customer edge routers. In a QoS congestion event, the PE node's priority queuing algorithm, which allocates bandwidth and delay guarantees to various traffic classes, fails to manage the available bandwidth efficiently. This leads to microbursts of high-priority traffic, interface tail drops due to excessive congestion, and QoS queue starvation, resulting in packet loss and network performance degradation. The underlying cause of this issue is often related to misconfigured QoS policies, incorrect policing rates, or inadequate buffer sizes, which can be exacerbated by environmental factors such as high network loads or faulty hardware.

Pre-requisite Checks

- Check: Verify that the redundant link status is up and functioning properly., Action: Run the command `show redundancy status` to confirm.

- Check: Ensure that the network load is within the normal operating range., Action: Run the command `show interface stats` to monitor the interface statistics.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Incorrect policing rates	Medium
Inadequate buffer sizes	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy-map, Expected Output: Policy-map name, policy-map type, and policy-map status, Description: This command displays the list of policy maps configured on the node. It helps identify if there are any misconfigured QoS policies that may be contributing to the congestion event.

Command: show interface policing, Expected Output: Policing rate, policing burst size, and policing action, Description: This command displays the policing rates and burst sizes configured on the interfaces. It helps identify if the policing rates are set too low, leading to excessive congestion.
```

```
Command: show buffer statistics, Expected Output: Buffer size, buffer utilization, and buffer overflow count, Description: This command displays the buffer statistics, including the buffer size, utilization, and overflow count. It helps identify if the buffer sizes are inadequate, leading to QoS queue starvation.
```

Resolution Steps

Step 1: Step: Reconfigure QoS policies, Command: configure qos policy-map, Description: Reconfigure the QoS policies to ensure that they are properly set up to manage traffic classes and allocate bandwidth efficiently. This may involve adjusting policing rates, buffer sizes, or other QoS parameters.

Step 2: Step: Adjust policing rates, Command: configure policing, Description: Adjust the policing rates to ensure that they are set high enough to prevent excessive congestion. This may involve increasing the policing rate or adjusting the policing burst size.

Step 3: Step: Increase buffer sizes, Command: configure buffer size, Description: Increase the buffer sizes to ensure that they are adequate to prevent QoS queue starvation. This may involve increasing the buffer size or adjusting the buffer allocation algorithm.

Post-Check Verification

```
Command: show qos policy-map, Expected Output: Policy-map name, policy-map type, and policy-map status, Description: Verify that the QoS policies are properly configured and are not contributing to the congestion event.

Command: show interface policing, Expected Output: Policing rate, policing burst size, and policing action, Description: Verify that the policing rates are set high enough to prevent excessive congestion.

Command: show buffer statistics, Expected Output: Buffer size, buffer utilization, and buffer overflow count, Description: Verify that the buffer sizes are adequate to prevent QoS queue starvation.
```

Roll-back Procedures

- Step: Restore original QoS policies, Command: configure qos policy-map original, Description: Restore the original QoS policies to their previous state before making any changes.

- Step: Reset policing rates, Command: configure policing original, Description: Reset the policing rates to their original values before making any changes.

- Step: Reset buffer sizes, Command: configure buffer size original, Description: Reset the buffer sizes to their original values before making any changes.